

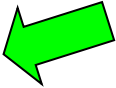
Frequent Patterns

Big Data Analytics

Readings

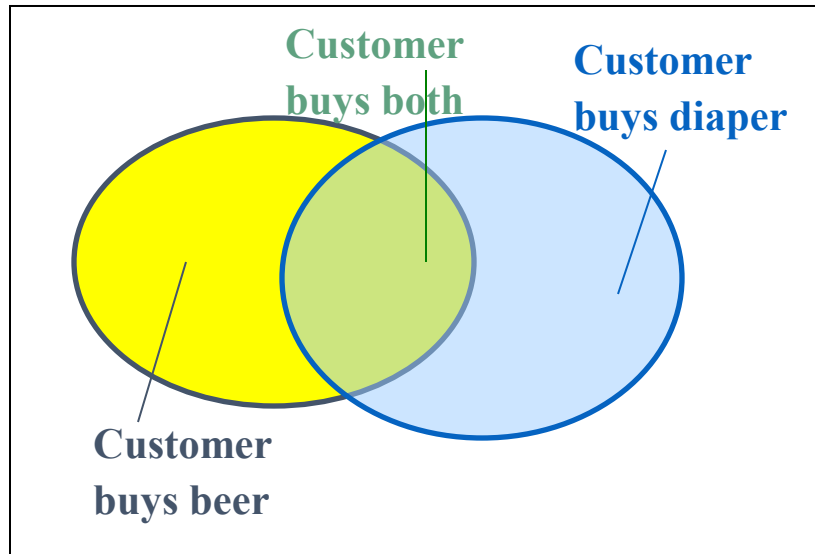
- “Data Mining – Concepts and Techniques” by Jiawei Han, Micheline Kamber, Jian Pei, third edition
 - Chap. 6

Chapter 6: Mining Frequent Patterns, Association and Correlations: Basic Concepts and Methods

- Basic Concepts 
- Frequent Itemset Mining Methods
- Which Patterns Are Interesting?—Pattern Evaluation Methods
- Summary

Basic Concepts: Frequent Patterns

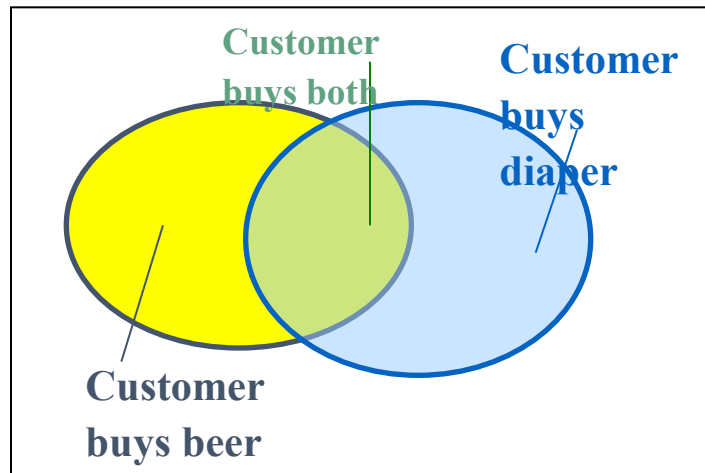
Tid	Items bought
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk



- **itemset**: A set of one or more items
- **k-itemset** $X = \{x_1, \dots, x_k\}$
- **(absolute) support**, or, **support count** of X : Frequency or occurrence of an itemset X
- **(relative) support**, s , is the fraction of transactions that contains X (i.e., the probability that a transaction contains X)
- An itemset X is **frequent** if X 's support is no less than a *minsup* threshold

Basic Concepts: Association Rules

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- Find all the rules $X \rightarrow Y$ with minimum support and confidence
 - support**, s , probability that a transaction contains $X \cup Y$
 - confidence**, c , conditional probability that a transaction having X also contains Y

Let $minsup = 50\%$, $minconf = 50\%$

Freq. Pat.: Beer:3, Nuts:3, Diaper:4, Eggs:3,
{Beer, Diaper}:3

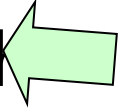
- Association rules: (many more!)
 - $Beer \rightarrow Diaper$ (60%, 100%)
 - $Diaper \rightarrow Beer$ (60%, 75%)

Computational Complexity of Frequent Itemset Mining

- How many itemsets are potentially to be generated in the worst case?
 - The number of frequent itemsets to be generated is sensitive to the minsup threshold
 - When minsup is low, there exist potentially an exponential number of frequent itemsets

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Scalable Frequent Itemset Mining Methods

- Apriori: A Candidate Generation-and-Test Approach 
- Improving the Efficiency of Apriori
- FPGrowth: A Frequent Pattern-Growth Approach
- ECLAT: Frequent Pattern Mining with Vertical Data Format

The Downward Closure Property and Scalable Mining Methods - Apriori

- The **downward closure** property of frequent patterns
 - Any subset of a frequent itemset must be frequent
 - If **{beer, diaper, nuts}** is frequent, so is **{beer, diaper}**
 - i.e., every transaction having {beer, diaper, nuts} also contains {beer, diaper}

TID	Items
1	Beer, Diaper, Eggs
2	Beer, Diaper, Nuts
3	Diaper, Nuts, Eggs, Coffee
4	Beer, Diaper, Nuts, Eggs, Coffee, Milk

Apriori: A Candidate Generation & Test Approach

- Apriori pruning principle: If there is **any** itemset which is infrequent, its superset should not be generated/tested! (Agrawal & Srikant @VLDB'94, Mannila, et al. @ KDD' 94)
- Method:
 - Initially, scan DB once to get frequent 1-itemset
 - {Beer}, {Nuts}, {Diaper}, {Eggs}

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 - Generate length $(k+1)$ candidate itemsets from length k frequent itemsets

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 - {Beer, Nuts}, {Beer, Diaper}, {Beer, Eggs}

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 - {Beer, Nuts}, {Beer, Diaper}, {Beer, Eggs}, {Nuts, Diaper}, {Nuts, Eggs}, {Diaper, Eggs}
 - Test the candidates against DB

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- Method:
 - Initially, scan DB once to get frequent 1-itemset
 - {Beer}, {Nuts}, {Diaper}, {Eggs}
 - Generate length (k+1) candidate itemsets from length k frequent itemsets
 - {Beer, Nuts}, {Beer, Diaper}, {Beer, Eggs}, {Nuts, Diaper}, {Nuts, Eggs}, {Diaper, Eggs}
 - Test the candidates against DB
 - Terminate when no frequent or candidate set can be generated

Implementation of Apriori

- How to generate candidates?
 - Step 1: self-joining L_k
 - Step 2: pruning
- Example of Candidate-generation
 - $L_3 = \{abc, abd, acd, ace, bcd\}$
 - Self-joining: $L_3 * L_3$
 - $abcd$ from abc and abd
 - $acde$ from acd and ace
 - Pruning:
 - $acde$ is removed because ade is not in L_3
 - $C_4 = \{abcd\}$

The Apriori Algorithm—Another Example with $\text{Sup}_{\min} = 2$

Transactional Data for an *AllElectronics* Branch

<i>TID</i>	<i>List of item_IDs</i>
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
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Scan *D* for
count of each
candidate
→

C_1	
Itemset	Sup. count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

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Compare candidate
support count with
minimum support
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Scan *D* for
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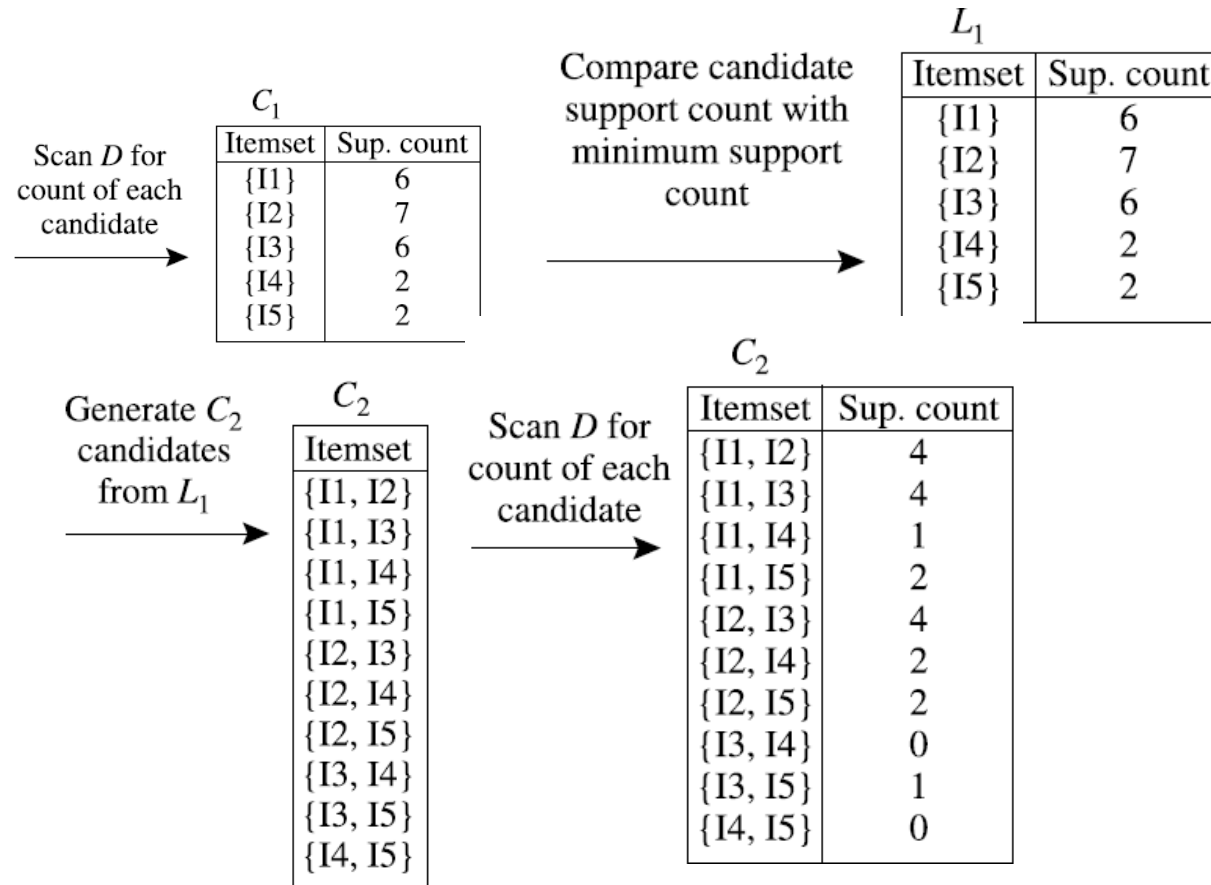
Generate C_2
candidates
from L_1

C_2	
Itemset	
{I1, I2}	
{I1, I3}	
{I1, I4}	
{I1, I5}	
{I2, I3}	
{I2, I4}	
{I2, I5}	
{I3, I4}	
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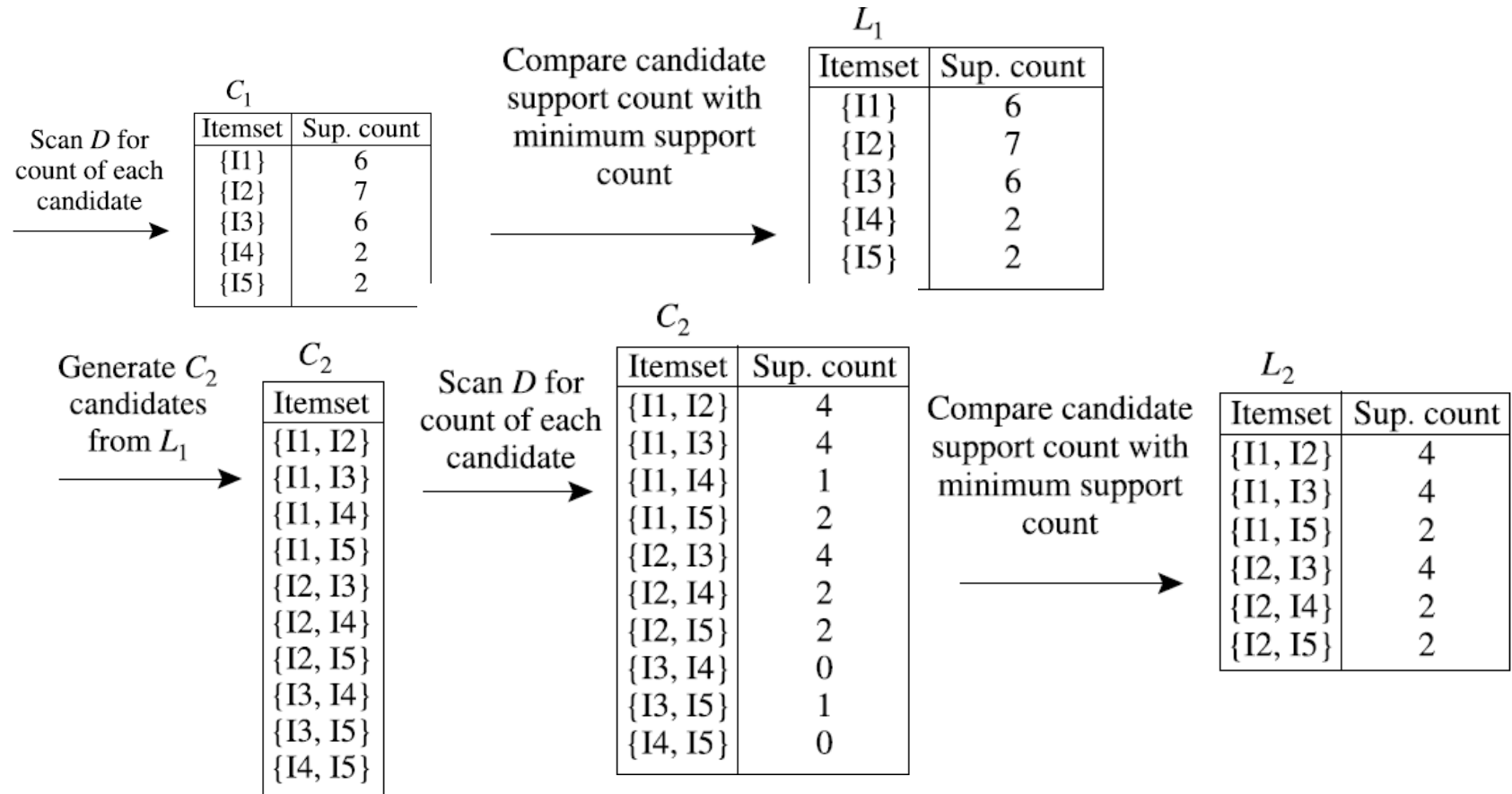
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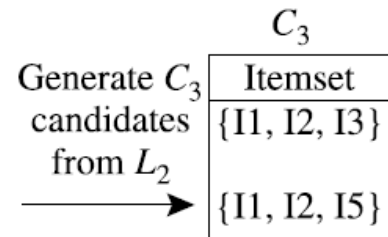
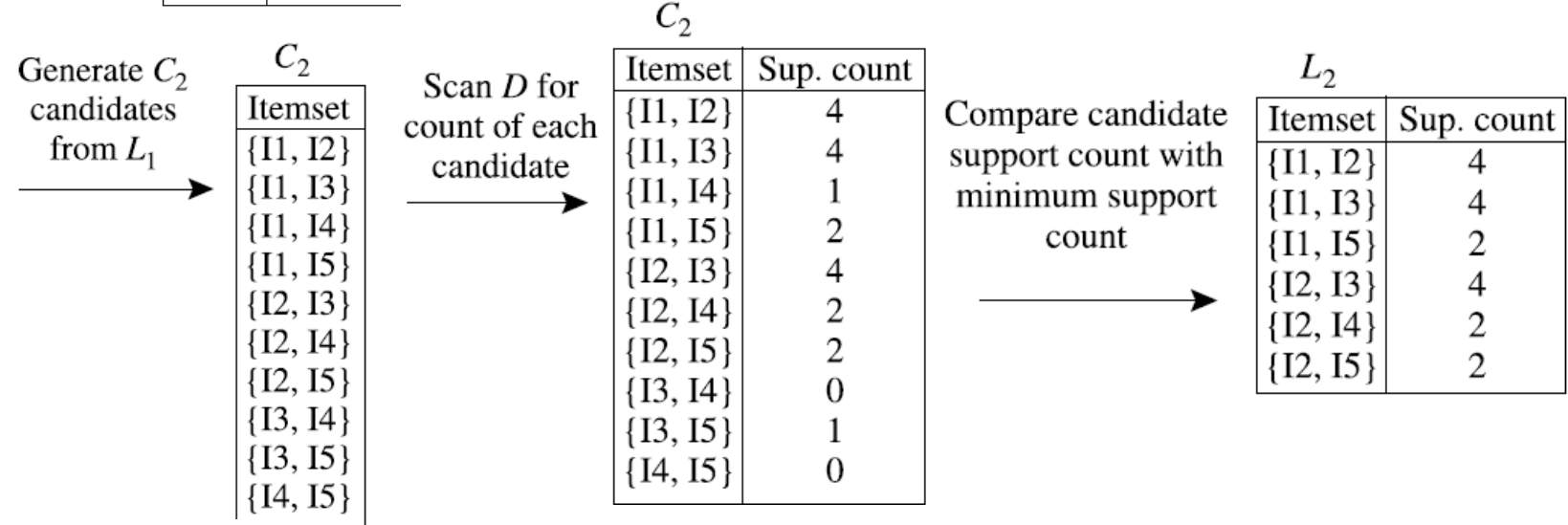
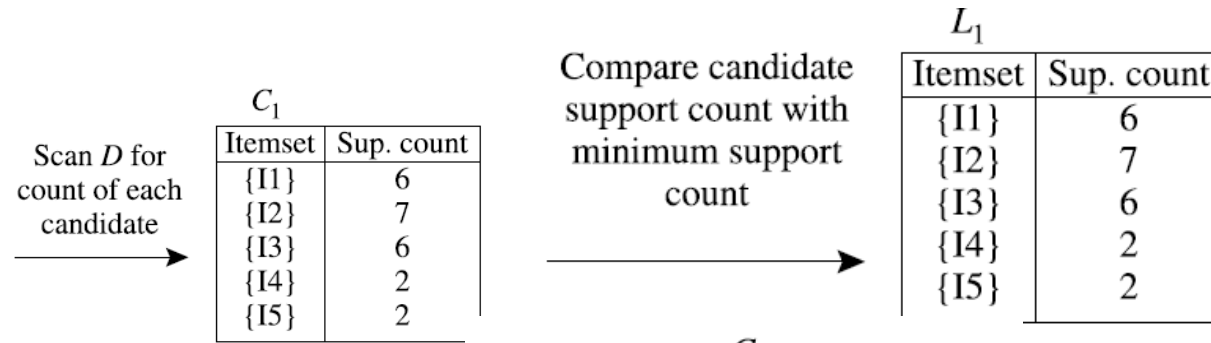
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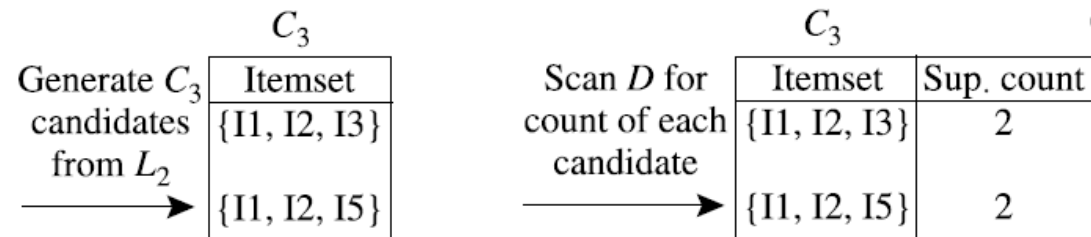
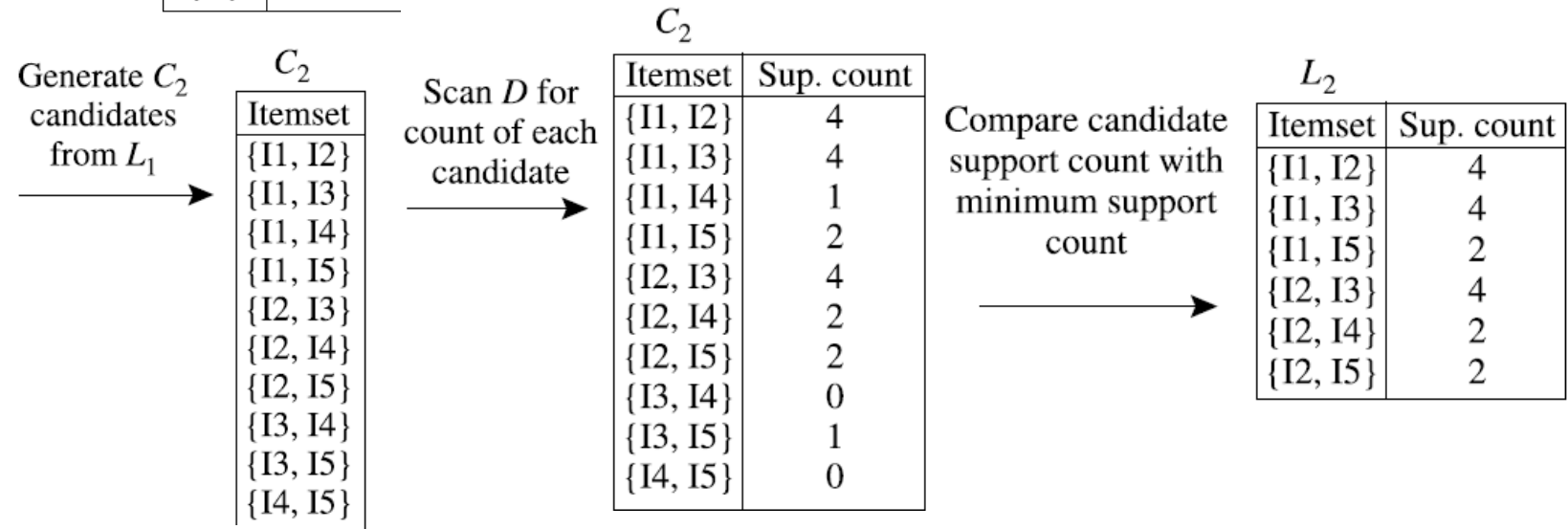
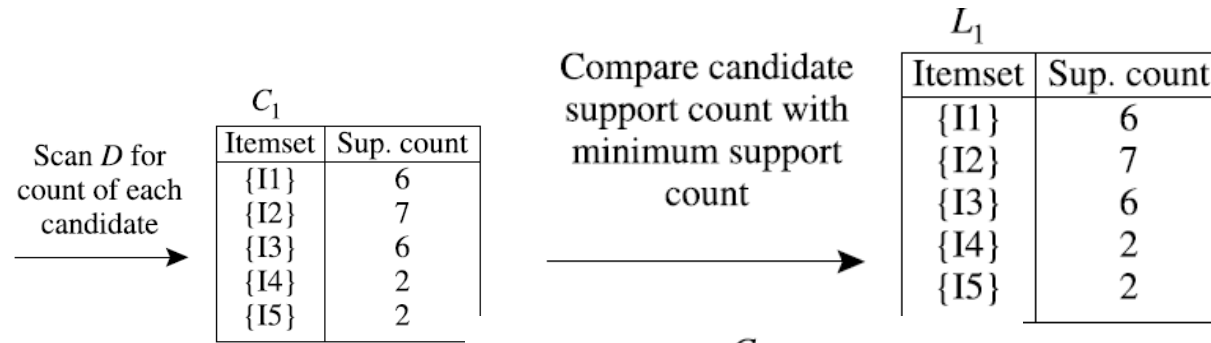
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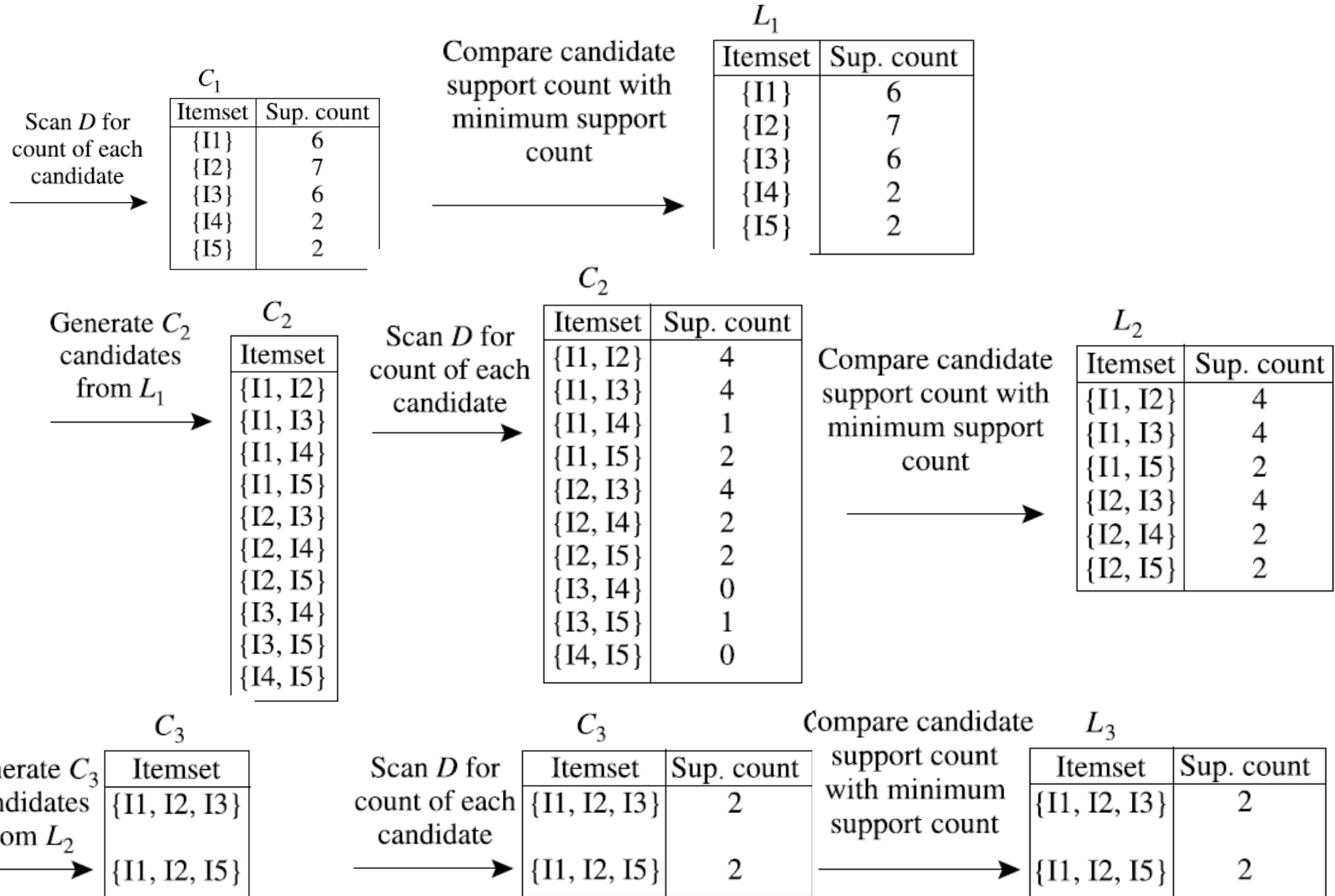
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Parallelize?

- Frequent 1 itemsets

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- generating candidate "k+1-itemset" (self-join and pruning)

- Frequent 1 itemsets
- generating candidate "k+1-itemset" (self-join and pruning)
- Generating frequent "k+1-itemsets" (consult DB)

Frequent 1 itemsets

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Generating candidate "k+1-itemset" (self-join and pruning)

$$L_3 = \{abc, abd, acd, ace, bcd\}$$

Generating frequent “ $k+1$ -itemsets” (consult DB)

- Frequent 1 itemsets
 - Word count with combine
- Self-join and pruning – generating “k+1-itemset” candidates
 - Map
 - Frequent K-itemsets are broadcasted
 - Input: <“self-join”, (partial k-itemsets)>
 - Self-join each k-itemset with broadcasted k-itemsets
 - Output: <“k+1-itemset”, 1>
 - Reduce
 - Frequent K-itemsets are broadcasted
 - Input: <“k+1-itemset”, “list of 1’s”>
 - Prune based on broadcasted k-itemsets
 - Output: <“candidate k+1-itemset”, 1>
- Generating frequent “k+1-itemsets”
 - Map
 - Input: <“candidate”, “list of candidate k+1-itemsets”>
 - Read partial DB
 - Output: <“k+1-itemset”, “no of occurrences in partial DB”>
 - Reduce
 - Input: <“k+1-itemset”, “list of no of occurrences in partial DB”>
 - Output: <“k+1-itemset”, “addition of occurrences”>

Applications of Frequent Pattern Analysis

- Basket data analysis

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- Basket data analysis
- Web log (click stream) analysis

Questions to ask

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- Number of transactions?

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- Number of frequent 2-itemsets?
- What's the longest frequent itemset?

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- Number of total frequent itemsets?
- Number of total rules?

Questions to ask

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- Number of total rules?
- Confidence?

Questions to ask

- Number of transactions?
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- Number of frequent 2-itemsets?
- What's the longest frequent itemset?
- Number of total frequent itemsets?
- Number of total rules?
- Confidence?
- How long does it take to compute all the frequent itemsets?

Classification

Big Data Analytics

Readings

- “Data Mining – Concepts and Techniques” by Jiawei Han, Micheline Kamber, Jian Pei, third edition
 - Chap. 8

Chapter 8. Classification: Basic Concepts

- Classification: Basic Concepts 
- Decision Tree Induction
- Bayes Classification Methods
- Rule-Based Classification
- Model Evaluation and Selection
- Techniques to Improve Classification Accuracy:
Ensemble Methods
- Summary

Supervised vs. Unsupervised Learning

- Supervised learning (classification)
 - Supervision: The training data (observations, measurements, etc.) are accompanied by **labels** indicating the class of the observations
 - New data is classified based on the training set
- Unsupervised learning (clustering)
 - The class labels of training data is unknown
 - Given a set of measurements, observations, etc. with the aim of establishing the existence of classes or clusters in the data

Prediction Problems: Classification vs. Numeric Prediction

- Classification

- predicts categorical class labels (discrete or nominal)
- classifies data (constructs a model) based on the training set and the values (**class labels**) in a classifying attribute and uses it in classifying new data

- Numeric Prediction

- models continuous-valued functions, i.e., predicts unknown or missing values

- Typical applications

- Credit/loan approval:
- Medical diagnosis: if a tumor is cancerous or benign
- Fraud detection: if a transaction is fraudulent
- Web page categorization: which category it is